

MALLAREDDY TECHNICAL CAMPUS.

SHORTS.

UNIT-I Short Answer Questions.

①

write 3-bit binary table?

Sol:-

Three Bit binary table:-

	4	2	1
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

②

Convert $(011110011001)_2$ to octal equivalent?

Sol:-

Given that, $(011110011001)_2$

now, $(011110011001)_2 \rightarrow (?)_8$

now, $011 \mid 110 \mid 011 \mid 001$
 3 6 3 1

$\therefore (011110011001)_2 \rightarrow (3631)_8$

octal table.

	4	2	1
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

③

write 4-bit binary table?

Sol:-

Four bit binary table:-

	8	4	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10(A)	1	0	1	0
11(B)	1	0	1	1
12(C)	1	1	0	0
13(D)	1	1	0	1
14(E)	1	1	1	0
15(F)	1	1	1	1

④

Convert $(111110011)_2$ to hexadecimal equivalent?

Sol:-

Given that, $(111110011)_2$

now, $(111110011)_2 \longrightarrow (?)_{16}$

Now, $0001 \mid 1111 \mid 0011$
 $1 \quad F \quad 3$

$$\therefore (111110011)_2 \longrightarrow (1F3)_{16}$$

Hexadecimal table.

	8	4	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10 (A)	1	0	1	0
11 (B)	1	0	1	1
12 (C)	1	1	0	0
13 (D)	1	1	0	1
14 (E)	1	1	1	0
15 (F)	1	1	1	1

⑤ Convert $(643)_8$ to binary equivalent?

Sol:-

$$\text{Now, } (643)_8 \longrightarrow (?)_2$$

$$\text{Now, } (643)_8$$

$$\text{Now, } \begin{array}{c|c|c} 6 & 4 & 3 \\ \hline 110 & 100 & 011 \end{array}$$

$$\therefore (643)_8 \longrightarrow (110100011)_2$$

Octal table.

	4	2	1
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

⑥ Explain "AND" logical operator?

Sol:

AND:- The "AND" operator is a binary operator, denoted as " $\wedge$ " (or) sometimes by just concatenating symbols is true only if both parameters are true.

Truth table:-

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

The expression A and B is often written as " $A \wedge B$ " (or) " $A \& B$ " (or) " $A \cdot B$ ". The AND operator is known as a conjunction. " $A \wedge B$ " and " AB " are the most common notations.

⑦ Convert $(1010110)_2$ to Hexadecimal form?

Sol:-

Given that, $(1010110)_2$

Now, $(1010110)_2 \longrightarrow (?)_{16}$

Now, $0101 \mid 0110$
5 6

$\therefore (1010110)_2 \longrightarrow (01010156)_{16}$

Hexadecimal table.

	8	4	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10(A)	1	0	1	0
11(B)	1	0	1	1
12(C)	1	1	0	0
13(D)	1	1	0	1
14(E)	1	1	1	0
15(F)	1	1	1	1

⑧ Convert $(B3)_{16}$ to binary?

Sol:-

Now, $(B3)_{16} \longrightarrow (?)_2$

Now, B | 3
 1011 | 0011

$\therefore (B3)_{16} \longrightarrow (10110011)_2$

Hexadecimal table.

	8	4	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10(A)	1	0	1	0
11(B)	1	0	1	1
12(C)	1	1	0	0
13(D)	1	1	0	1
14(E)	1	1	1	0
15(F)	1	1	1	1

UNIT-I Long Answer Questions.

①

Convert $(11001)_2$ to decimal conversion?

Sol:

Given that, $(11001)_2 \rightarrow (?)_{10}$.

Formula: $N_b = x_{n-1} b^{n-1} + x_{n-2} b^{n-2} + \dots \rightarrow \text{①}$

Given, $n=5, b=2$.

Substituting n and b values in equation ①, we can get

$$N_b = x_{5-1} \cdot 2^{5-1} + x_{5-2} \cdot 2^{5-2} + x_{5-3} \cdot 2^{5-3} + x_{5-4} \cdot 2^{5-4} + x_{5-5} \cdot 2^{5-5}$$

$$= x_4 \cdot 2^4 + x_3 \cdot 2^3 + x_2 \cdot 2^2 + x_1 \cdot 2^1 + x_0 \cdot 2^0$$

$$= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 16 + 8 + 0 + 0 + 1 = 25$$

$$\therefore (11001)_2 \rightarrow (25)_{10}$$

$$\left[\begin{array}{c} 11001 \\ \downarrow \downarrow \downarrow \downarrow \downarrow \\ x_4 x_3 x_2 x_1 x_0 \end{array} \right]$$

$$[\because 2^0 = 1]$$

$$[\because 2^0 = 1]$$

②

Explain logical operators with truth tables?

Sol:

Logical operations:-

① AND:- The "AND" operator is a binary operator, denoted as " $\wedge$ ", "&" or sometimes by just concatenating symbols

is true only if both parameters are true.

Truth table:-

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

The expression "A and B" is often written as " $A \wedge B$ " (or " $A \& B$ " (or) " $A \cdot B$ "). The "AND" operator is also known as a conjunction.

" $A \wedge B$ " and " $A \& B$ " are the most common notations.

2. OR:- The "or" operator is a binary operator, denoted as ' $\vee$ ', (or) + is true if either parameter is true.

Truth table:-

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

The expression " $A \times B$ " is often written as " $A \vee B$ ".
The ^{or}operator is also known as disjunction.

3) Not:- The "Not" operator is a unary operator denoted as " $\neg$ ", " $!$ ", " $-$ ", or as a line overtop the parameter is true, if the parameter is false.

Truth table:-

A	$\neg A$
T	F
F	T

The expression " $\text{NOT } A$ " is often written as " $\neg A$ ". The "Not" operator is also known as negation.

4) Exclusive-or (XOR):- The Exclusive-or operator is binary operator, denoted as $\oplus$.

Truth table:-

A	B	$A \oplus B$
T	T	F
T	F	T
F	T	T
F	F	F

Expression " $A \times B$ " is often written " $A \oplus B$ ". Latex notation is normally only seen in the context of C-derived programming languages such as C++ (or) Java.

5) NAND:- the "NAND" operator is a binary operator denoted as " $\uparrow$ " (or) " $\bar{\wedge}$ " as an and combined with a not, is false only when both parameters are true.

Truth table:-

A	B	$A \wedge B$	$\neg(A \wedge B)$ (or) $A \uparrow B$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

The expression "A and B" is often written " $A \uparrow B$ " (or) " $\neg(A \wedge B)$ ".

6) NOR:- the "NOR" operator is binary operator, denoted as " $\downarrow$ " (or) " $\bar{\vee}$ " as an or combined with a not, is true only when both parameters are false.

Truth table:-

A	B	$A \vee B$	$\neg(A \vee B)$ (or) $A \downarrow B$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T

The expression " $A \text{ nor } B$ " is often written " $\neg(A \vee B)$ ".

7) Conditional (or) Implication (or) Implies:- $(\Rightarrow \text{ or } \rightarrow)$:-

Let A and B be two propositions the Implication " $p \rightarrow q$ " is the proposition that is false when A is true and B is false otherwise True.

Truth table:-

A	B	$A \rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

8) BiConditional (or) Bi-Implication:- (If and only if) (or) $(\leftrightarrow)$:-

Let A and B be two propositions, the biconditional statement $A \leftrightarrow B$ is the proposition "A if and only if B" the Bi-conditional statement $A \leftrightarrow B$ is true when A and B have same truth values otherwise false.

Truth table:-

A	B	$A \leftrightarrow B$
T	T	T
T	F	F
F	T	F
F	F	T

③ Explain bitwise operators with truth tables?

Ans Bit wise operators:- Bitwise operators perform on bit level is also called "Bit manipulation" operators.

① AND "&":-

Truth table:-

AND "&"		
Input-1	Input-2	Output
0	0	0
0	1	0
1	0	0
1	1	1

② OR "1" :-

Truth table :-

OR "1"		
Input-1	Input-2	outPut
0	0	0
1	0	1
0	1	1
1	1	1

③ NAND "& ~" :-

Truth table :-

NAND "& ~"		
Input-1	Input-2	outPut
0	0	1
0	1	1
1	0	1
1	1	0

④

Find 2's Complement of $(1001)_2$?

Sol

Given, $1001 \rightarrow$ normal number.

$0110 \rightarrow$ 1's Complement.

Find out 2's Complement:- 1's Complement + 1.

Now, 1's Complement = 0110

2's Complement = $\begin{array}{r} 0110 \\ +1 \\ \hline 0111 \end{array}$

$\therefore$ 2's Complement = 0111

⑤ Represent +45 and -45 using 16 bits

(i) sign-magnitude form.

(ii) 1's Complement form.

(iii) 2's Complement form.

Sol

Given that, the normal number +45.
To find +45 using 16 bits

(i) sign magnitude form:-

MSB
+45 = $\begin{array}{|c|c|c|c|c|c|c|c|c|c|c|c|c|c|c|c|} \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ \hline \end{array}$ LSB
signbit magnitude.

(ii) 1's Complement form:-

MSB
+45 = $\begin{array}{|c|c|c|c|c|c|c|c|c|c|c|c|c|c|c|c|} \hline 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ \hline \end{array}$ LSB
signbit magnitude.

(iii) 2's Complement form:-

$$2's \text{ Complement} = 1's \text{ Complement} + 1$$

$$+45 = 011111111010010$$

+1

$$\begin{array}{c} \boxed{0}111111111010011 \rightarrow 2's \text{ Complement} \\ \swarrow \\ \text{signbit} \end{array}$$

To find -45 using 16 bits:-

Given that the normal number -45.

(i) sign magnitude form:-

$$-45 = \begin{array}{c} \text{MSB} \qquad \qquad \qquad \text{LSB} \\ \boxed{1} \mid 0000000000101101 \\ \swarrow \qquad \qquad \qquad \underbrace{\hspace{10em}} \\ \text{signbit} \qquad \qquad \text{magnitude} \end{array}$$

(ii) 1's Complement form:-

$$-45 = \begin{array}{c} \text{MSB} \qquad \qquad \qquad \text{LSB} \\ \boxed{1} \mid 111111111010010 \\ \swarrow \qquad \qquad \qquad \underbrace{\hspace{10em}} \\ \text{signbit} \qquad \qquad \text{Magnitude} \end{array}$$

(iii) 2's Complement form:-

$$2's \text{ Complement} = 1's \text{ Complement} + 1$$

$$\text{Now, } -45 = 0000000000101101 = +45$$

$$\underline{\underline{1's \text{ Complement:-}}} \quad 1111111111010010$$

$$2's \text{ Complement} = \begin{array}{r} 1111111111010010 \\ +1 \\ \hline 1111111111010011 \end{array}$$

⑥ Convert $(7365)_8$ to its equivalent hexadecimal number?

Sol:-

$$\text{Given that } (7365)_8 \rightarrow (?)_{16}$$

Step-I:- Convert $(7365)_8$ into binary

$$\text{Now, } \begin{array}{c|c|c|c} 7 & 3 & 6 & 5 \\ \hline 111 & 011 & 110 & 101 \end{array}$$

$$\therefore (7365)_8 \rightarrow (111011110101)_2$$

Step-II:- Convert $(111011110101)_2$ into Hexadecimal

$$\text{Now, } \begin{array}{c|c|c} 1110 & 1111 & 0101 \\ \hline E & F & 5 \end{array}$$

$$\therefore (7365)_8 \rightarrow (EF5)_{16}$$

Octal table:-

	4	2	1
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Hexadecimal table:-

	8	4	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10(A)	1	0	1	0
11(B)	1	0	1	1
12(C)	1	1	0	0
13(D)	1	1	0	1
14(E)	1	1	1	0
15(F)	1	1	1	1

⑦ Express -45 in 2's Complement using 8 bit?

Sol:- Given that the normal number -45.

To find 2's Complement form:-

Now, 2's Complement = 1's Complement + 1.

Now, -45 = 00101001 = +45

1's Complement Form:-

11010010

+1

2's Complement =

11010011

⑧ If $A=4$ and $B=7$, Find out $A \& B$, A/B .

octal table.

21

Sol:-

Given that, $A=4$ and $B=7$.

Now, $A=4 =$

4 2 1
1 0 0

$B=7 =$

4 2 1
1 1 1

Find out $A \& B$:-

Now $A \& B =$

1 0 0
1 1 1

$\therefore A \& B =$ 1 0 0

Find out A/B :-

Now, $A/B =$

1 0 0
1 1 1

$\therefore A/B =$ 1 1 1

UNIT-II Short Answer Questions.

① Find the $\text{GCD}(15, 28, 50)$.

Sol:-

The divisors of 15 are $= 1, 3, 5, 15$.

The divisors of 28 are $= 1, 2, 4, 7, 14, 28$.

The divisors of 50 are $= 1, 2, 5, 10, 25, 50$.

The common divisors $= 1$.

$\therefore \text{GCD} = 1$

② Prove that the product of two even integers is an even integer?

Sol:- Let a and b are two even integers then we have,

$$a = 2q_1, \quad b = 2q_2 \quad \text{where } q_1, q_2 \in \mathbb{Z}.$$

Product:- Now, $a \cdot b = 2q_1 \cdot 2q_2$
 $= 4q_1q_2 \quad [\because q = q_1, q_2]$
 $= 4q.$

$\therefore a \cdot b$ are Even integers.

$\therefore$ The product of two even integers is an even integer.

③ Find the common divisors of 24 and 56.

Sol:-

The divisors of 24 are $= 1, 2, 3, 4, 6, 8, 12, 24$.

The divisors of 56 are $= 1, 2, 4, 7, 8, 14, 28, 56$.

$\therefore$ The common divisors $= 1, 2, 4, 8$.

④ prove that the sum of two even Integers is an even Integer?

Sol

let a and b are two even Integers, then we have

$$a = 2q_1 \text{ and } b = 2q_2 \text{ where } q_1, q_2 \in \mathbb{Z}$$

Sum:-

$$\text{Now, } a + b = 2q_1 + 2q_2$$

$$= 2(q_1 + q_2) \quad [\because q = q_1 + q_2]$$

$$= 2q$$

$$\therefore a + b = 2q$$

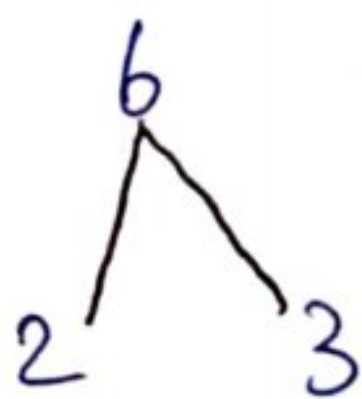
$\therefore a + b$ is Even Integer.

$\therefore$ the sum of two even Integers is an even Integer.

⑤ Find the LCM (6, 9)?

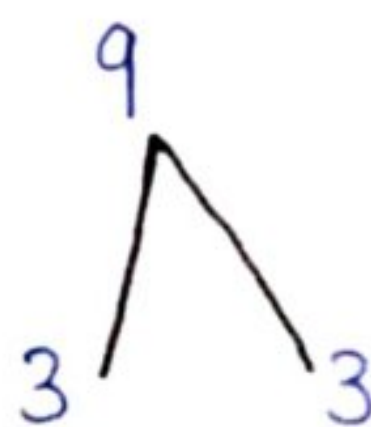
Sol:-

Now,



$$\therefore 6 = 2^1 \times 3^1$$

Now,



$$\therefore 9 = 3^1 \times 3^1 = 3^2$$

the LCM of prime factorization formula:-

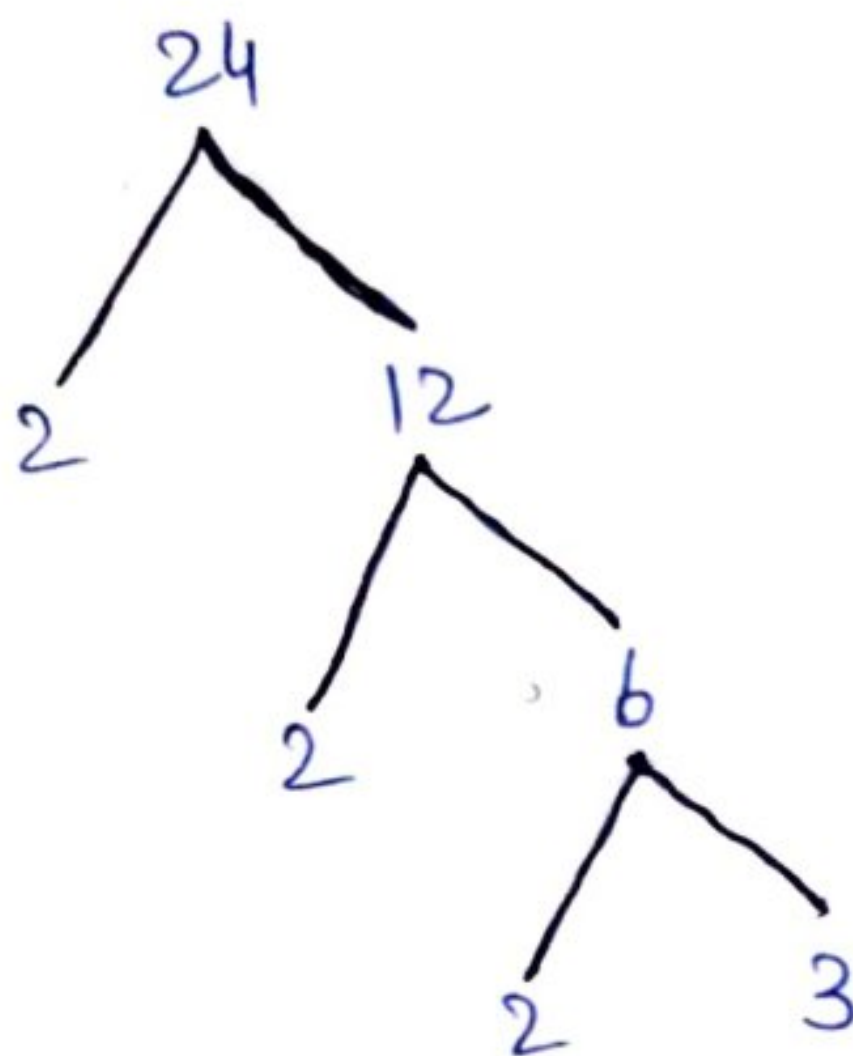
$$\begin{aligned} \text{LCM}(a, b) &= p_1^{\max(a, b)} \times p_2^{\max(a, b)} \times p_3^{\max(a, b)} \times \dots \\ &= 2^{\max(1, 0)} \times 3^{\max(1, 2)} \\ &= 2^1 \times 3^2 = 2 \times 9 = 18 \end{aligned}$$

⑥ Find the prime factorisation of 24?

Sol:-

Prime factorisation:-

Now,



$$\begin{aligned}\text{Now } 24 &= 2 \times 2 \times 2 \times 3 \\ &= 2^3 \times 3^1\end{aligned}$$

The prime factorisation Formula:-

$$P_1^{\alpha_1} \times P_2^{\alpha_2} \times \dots$$
$$2^3 \times 3^1$$

⑦ Find GCD (180, 48)?

Sol:-

The divisors of 180 are = 1, 2, 3, 4, 5, 6, 9, 10, 12, 15, 18, 20, 30, 36, 45, 60, 90, 180.

The divisors of 48 are = 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.

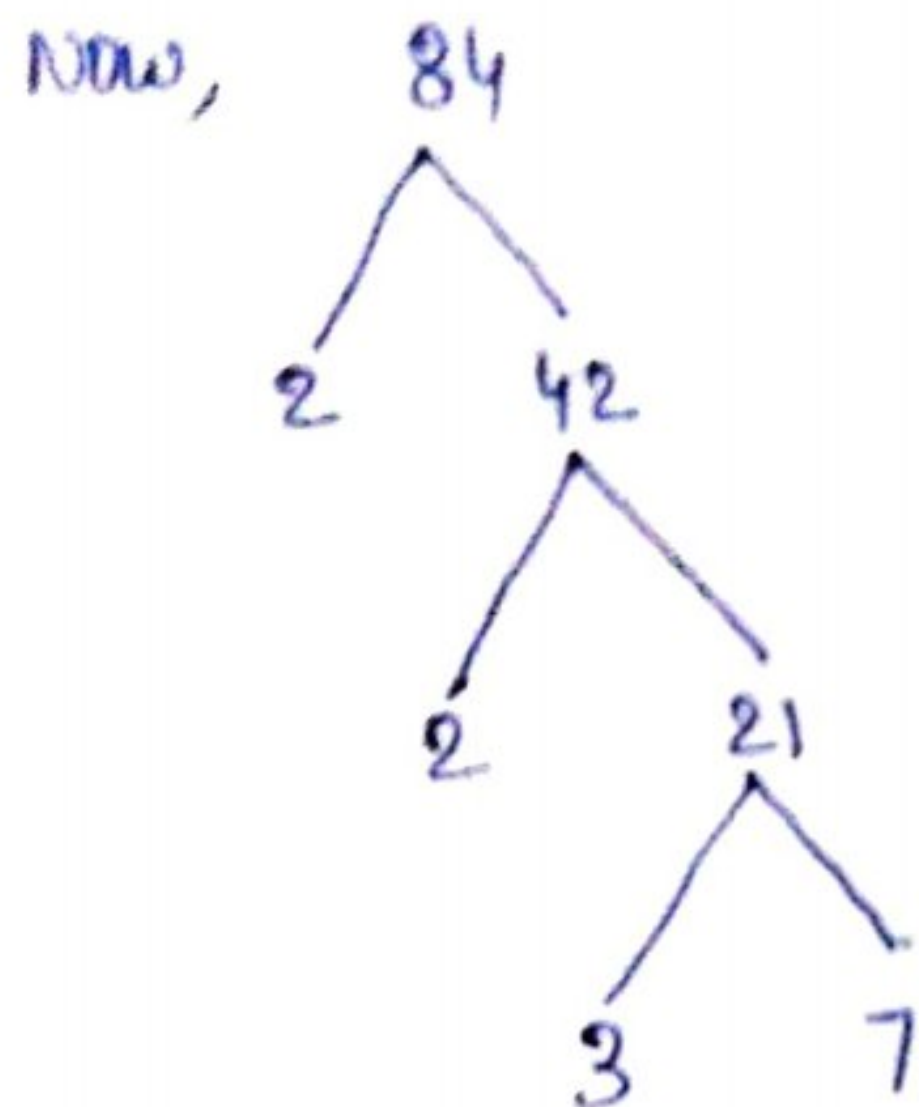
∴ The common divisors are = 1, 2, 3, 4, 6, 12,

∴ The GCD = 12.

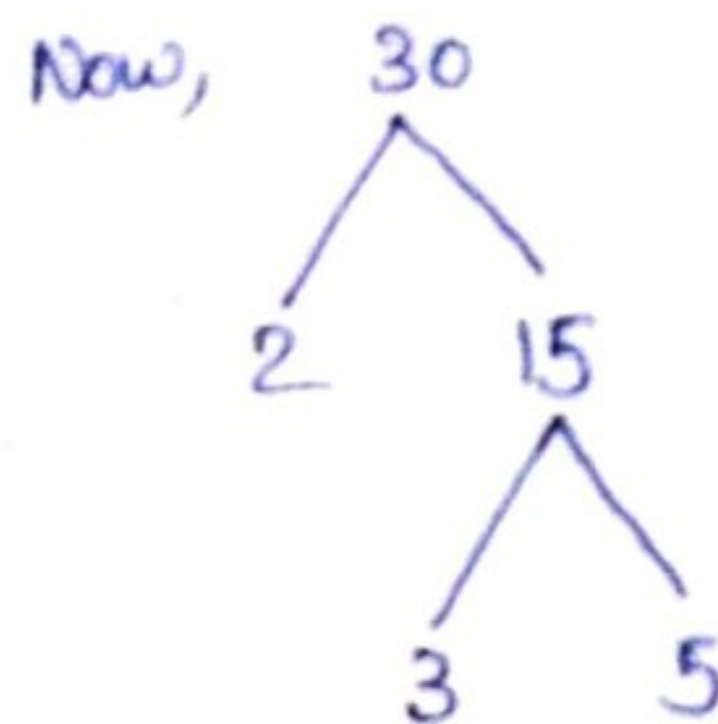
8

If $\text{GCD}(84, 30) = 6$ then $\text{LCM}(84, 30) = ?$

Sol:-



$$\begin{aligned}\text{Now, } 84 &= 2 \times 2 \times 3 \times 7 \\ &= 2^2 \times 3 \times 7\end{aligned}$$



$$\text{Now, } 30 = 2 \times 3 \times 5$$

The LCM of prime Factorization Formula:-

$$\begin{aligned}\text{LCM}(a, b) &= p_1^{\max(a, b)} \times p_2^{\max(a, b)} \times \dots \\ &= 2^{\max(2, 1)} \times 3^{\max(1, 1)} \times 5^{\max(1, 0)} \times 7^{\max(1, 0)} \\ &= 2^2 \times 3^1 \times 5^1 \times 7^1 \\ &= 4 \times 3 \times 5 \times 7 \\ &= 420\end{aligned}$$

UNIT-II

LONG ANSWER QUESTIONS

- ① Find the $\text{GCD}(-2210, 493)$ by using division Algorithm (or) Euclidean Algorithm?

Sol

Given that $\text{GCD}(-2210, 493)$.

Now Applying the theorem $\text{gcd}(a, b) = \text{gcd}(a, -b) = \text{gcd}(-a, b) = \text{gcd}(-a, -b) = \text{gcd}(|a|, |b|) = \text{gcd}(a, b)$.

$$\therefore \text{gcd}(-2210, 493) = \text{gcd}(|-2210|, 493) = \text{gcd}(2210, 493)$$

To find $\text{gcd}(2210, 493)$:-

$$\begin{array}{r} 493 \overline{) 2210} \quad (4 \\ \underline{1972} \\ 238 \end{array}$$
$$\begin{array}{r} 238 \overline{) 493} \quad (2 \\ \underline{476} \\ 17 \end{array}$$
$$\begin{array}{r} 17 \overline{) 238} \quad (14 \\ \underline{238} \\ 0 \end{array}$$

$$\therefore \text{gcd}(2210, 493) = 17 = d$$

To find Division Algorithm (or) Euclidean Algorithm:-

Now, by using division Algorithm (or) Euclidean Algorithm,

now,

$$2210 = 4 \times 493 + 238, \quad q_1 = 4, r_1 = 238$$

~~238~~

$$493 = 2 \times 238 + 17, \quad q_2 = 2, r_2 = 17$$

$$238 = 14 \times 17 + 0, \quad q_3 = 14, r_3 = 0.$$

Since $r_3 = 0$, $r_2 = 17$ is the last non-zero remainder.

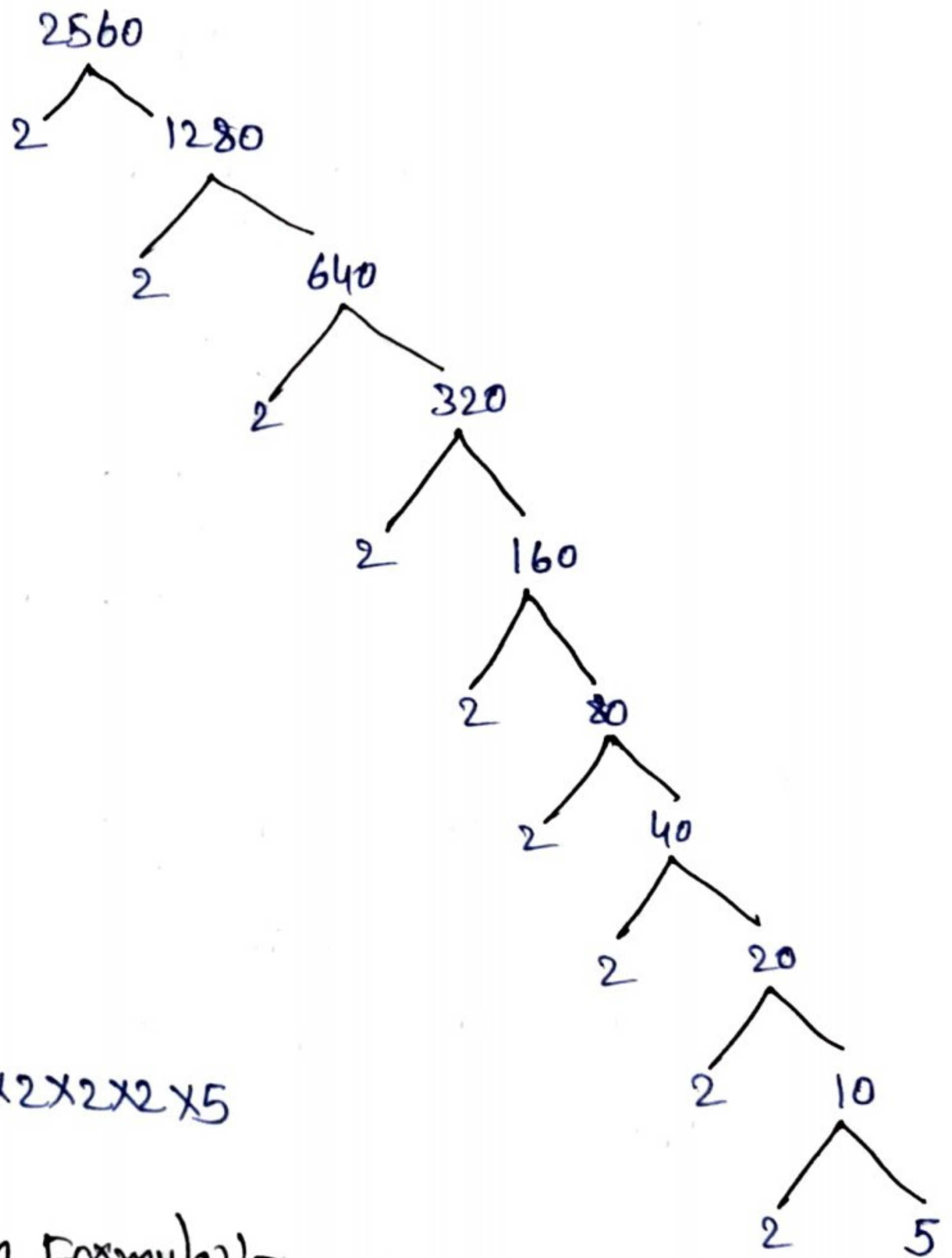
$$\therefore \gcd(2210, 493) = 17.$$

② Find the Prime Factorisation of 2560.

Soln

Prime Factorisation:- The representation of +ve Integers 'n' in the form $n = p_1^{\alpha_1} \times p_2^{\alpha_2} \times \dots \times p_x^{\alpha_x}$ is called Prime Factorisation of n is the canonical form.

Prime Factorisation:-



now,

$$2560 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 5$$

The Prime Factorisation Formula:-

$$p_1^{a_1} \times p_2^{a_2} \times p_3^{a_3} \times \dots$$

$$2560 = 2^9 \times 5^1$$

$$\therefore 2560 = 2^9 \times 5^1$$

③ Find the $\text{GCD}(45, 126)$ and express it in the form, $45x + 126y$ where x and y are integers?

Sol:-

First we compute $\text{gcd}(45, 126)$ by using Euclidean theorem.

$$\begin{array}{r} 45 \overline{) 126} \quad (2 \\ \underline{90} \\ 36 \overline{) 45} \quad (1 \\ \underline{36} \\ 9 \overline{) 36} \quad (4 \\ \underline{36} \\ 0 \end{array}$$

now,

$$126 = 2 \times 45 + 36, \quad q_1 = 2, \quad r_1 = 36$$

$$45 = 1 \times 36 + 9, \quad q_2 = 1, \quad r_2 = 9$$

$$36 = 4 \times 9 + 0, \quad q_3 = 4, \quad r_3 = 0$$

Since $r_3 = 0$, $r_2 = 9$ is the non-zero last remainder.

$$\therefore \text{gcd}(45, 126) = 9 = d$$

now, we find x and y such that, $d = ax + by$:-

$$\text{now, } d = ax + by$$

$$9 = 45x + 126y$$

To find x and y it is convenient to begin with the last non-zero remainder working backwards, substitution,

$$d = ax + by$$

$$9 = 45x + 126y.$$

Now,

$$\begin{aligned} 9 &= 45 - 1 \times 36 \\ &= 45 - 1 \times [126 - 2 \times 45] \\ &= 45 - 1 \times 126 + 2 \times 45 \\ &= 45[1 + 2] + 126(-1) \\ &= 45(3) + 126(-1). \end{aligned}$$

$$\therefore x = 3, y = -1.$$

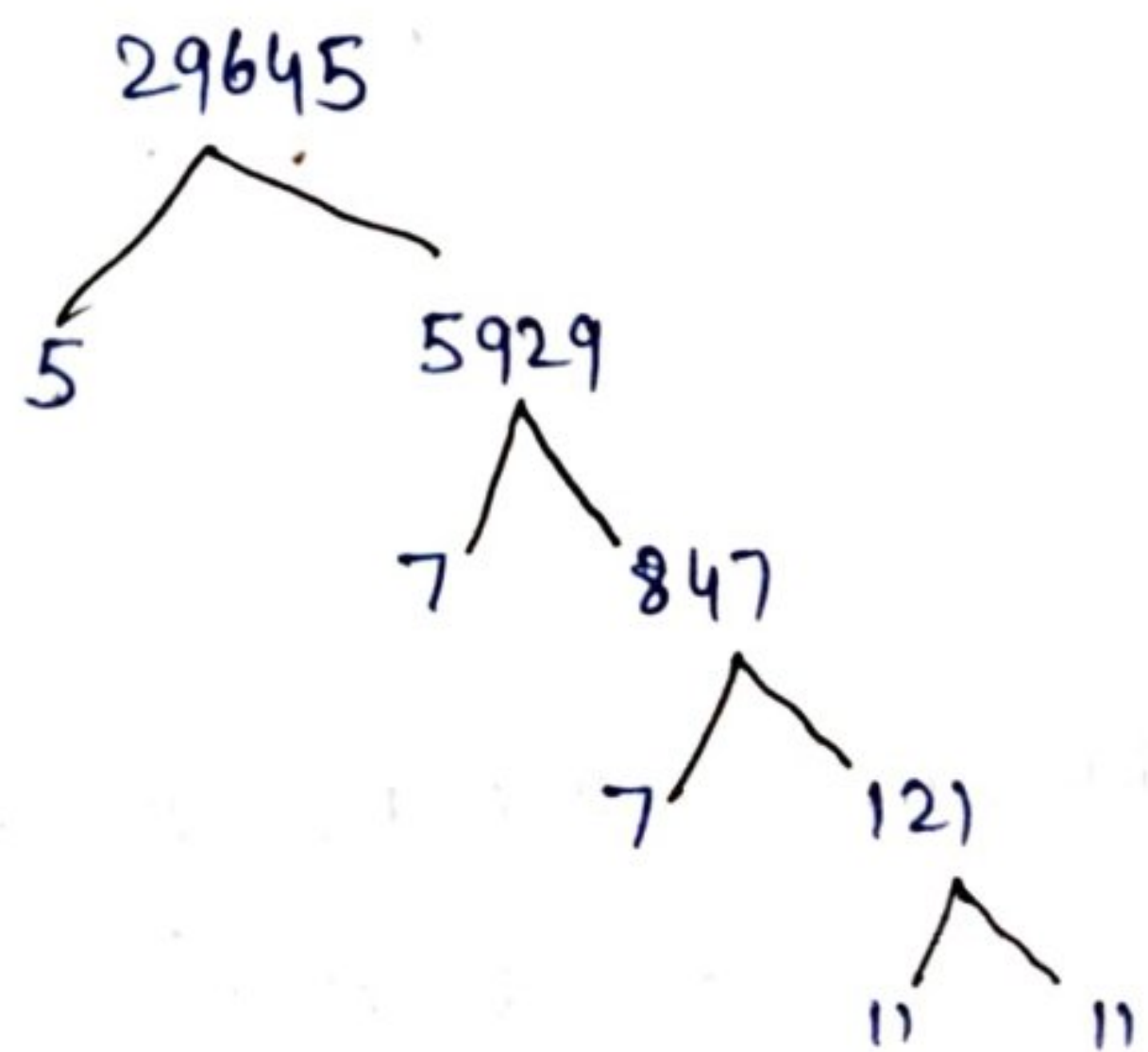
Q4

Find the prime factorisation of 29645.

Sol:

Prime Factorisation:- The representation of +ve integer 'n' in the form of $p_1^{a_1} \times p_2^{a_2} \times \dots \times p_r^{a_r}$ is called prime factorisation of 'n' canonical form.

Prime Factorisation:-



$$\text{Now, } 29645 = 5 \times 7 \times 7 \times 11 \times 11$$

The prime factorisation formula:-

$$\begin{aligned} &p_1^{a_1} \times p_2^{a_2} \times p_3^{a_3} \times \dots \\ &5^1 \times 7^2 \times 11^2. \end{aligned}$$

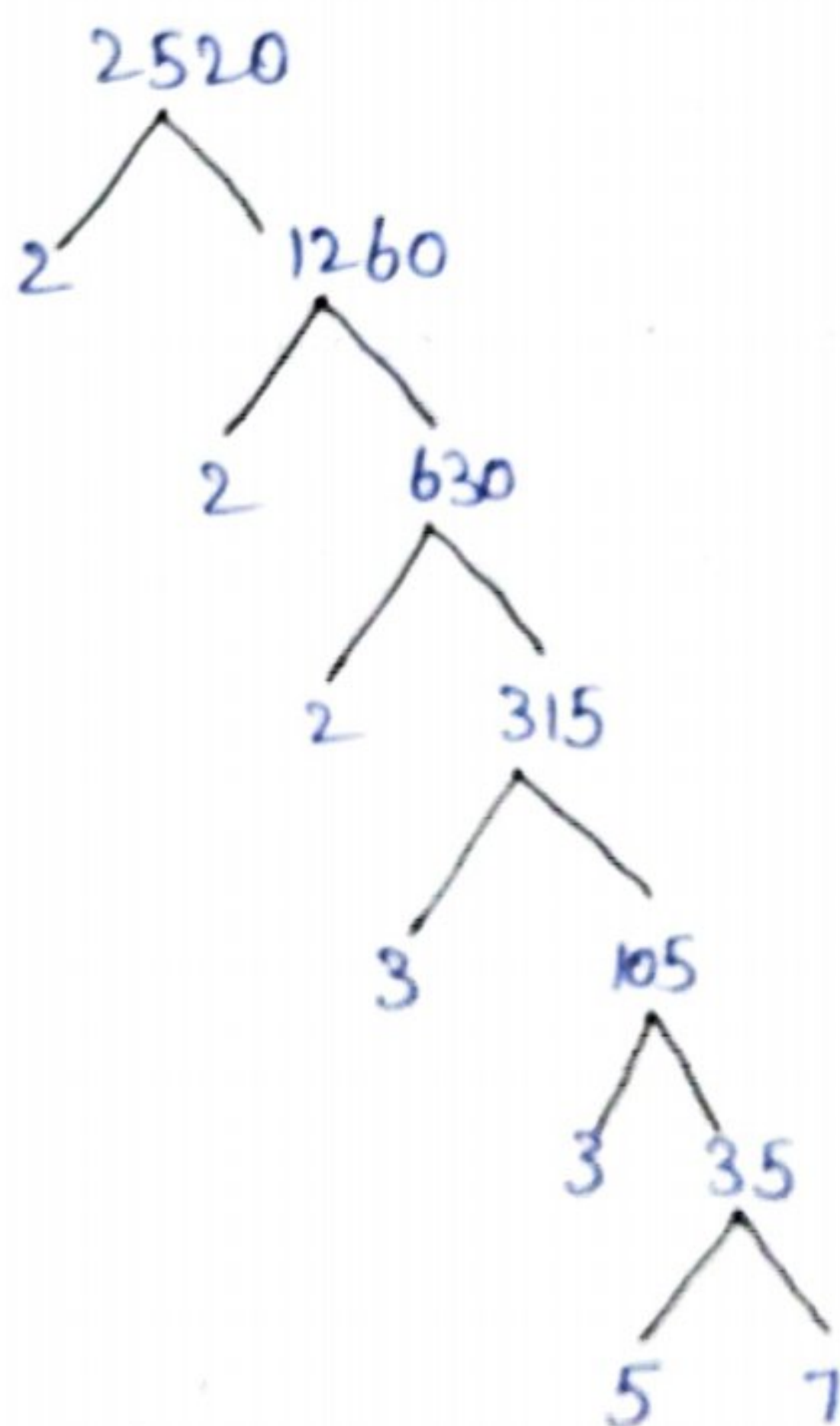
$$\therefore 29645 = 5^1 \times 7^2 \times 11^2$$

⑤ Find GCD and LCM of 2520 and 4950 using prime factorisation?

Sol:-

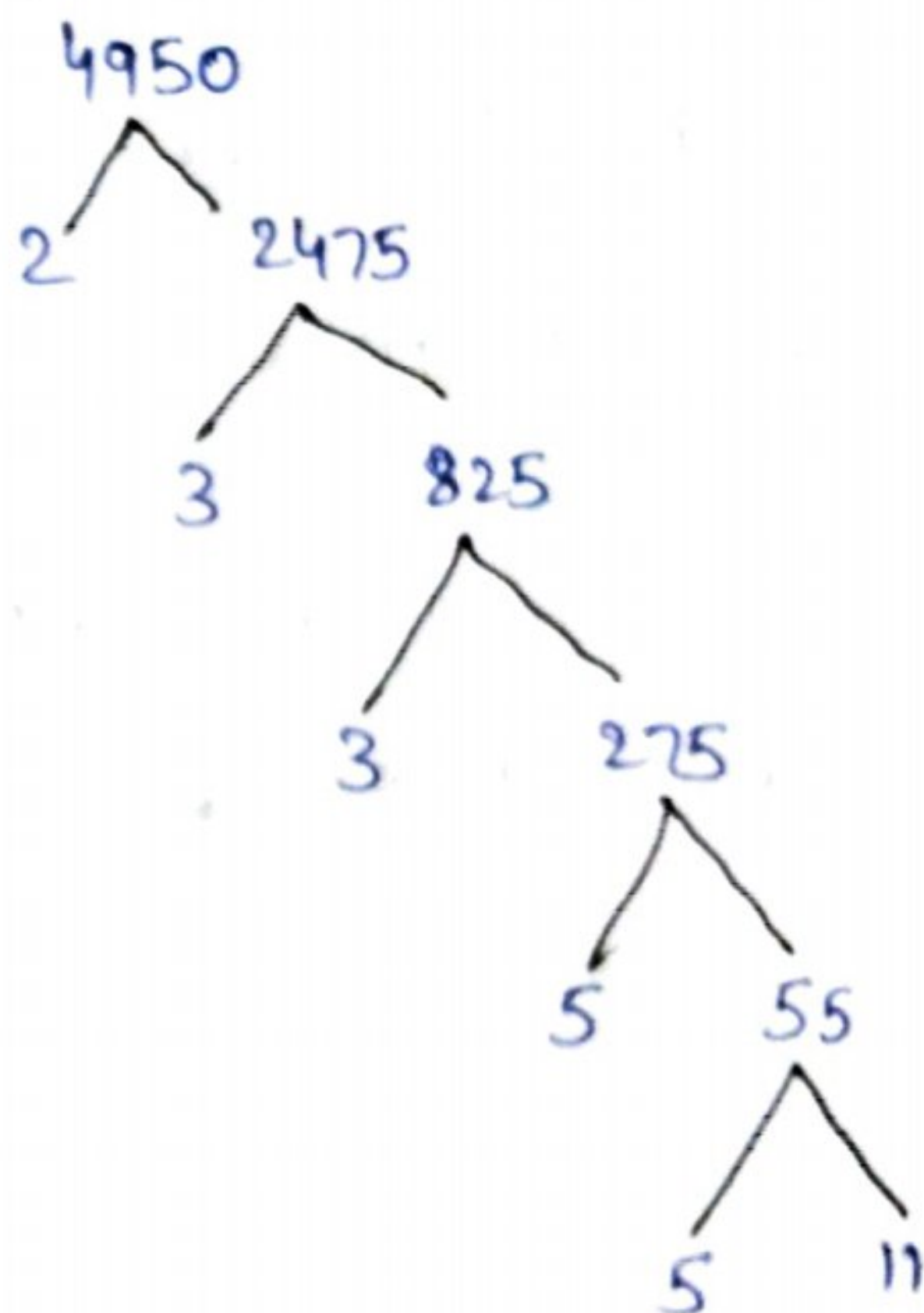
Prime Factorisation:- The representation of +ve Integers of n in the form of $n = p_1^{a_1} \cdot p_2^{a_2} \cdots p_x^{a_x}$ is called prime factorisation of n in the canonical form.

Prime Factorisation:-



$$\begin{aligned} \text{Now, } 2520 &= 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 7 \\ &= 2^3 \times 3^2 \times 5^1 \times 7^1 \end{aligned}$$

The Prime Factorisation Formula:-



$$\begin{aligned} \text{Now, } 4950 &= 2 \times 3 \times 3 \times 5 \times 5 \times 11 \\ &= 2^1 \times 3^2 \times 5^2 \times 11^1 \end{aligned}$$

$$p_1^{a_1} \times p_2^{a_2} \times \cdots$$

$$\text{Now, } 2520 = 2^3 \times 3^2 \times 5^1 \times 7^1$$

$$4950 = 2^1 \times 3^2 \times 5^2 \times 11^1$$

Prime Factorisation of GCD Formula:-

$$\text{GCD}(a, b) = p_1^{\min(a, b)} \times p_2^{\min(a, b)} \times p_3^{\min(a, b)} \times \dots$$

$$\text{GCD}(2520, 4950) = 2^{\min(3, 1)} \times 3^{\min(2, 2)} \times 5^{\min(1, 2)} \times 7^{\min(1, 0)} \times 11^{\min(0, 1)}$$

$$= 2^1 \times 3^2 \times 5^1 \times 7^0 \times 11^0$$

$$= 2^1 \times 3^2 \times 5^1 \times 1 \times 1$$

$$= 90$$

$$[\because 2^0 = 1]$$

$$[\because 7^0 = 1]$$

$$[\because 11^0 = 1]$$

Prime Factorisation of LCM Formula:-

$$\text{LCM}(a, b) = p_1^{\max(a, b)} \times p_2^{\max(a, b)} \times p_3^{\max(a, b)} \times \dots$$

$$\text{LCM}(2520, 4950) = 2^{\max(3, 1)} \times 3^{\max(2, 2)} \times 5^{\max(1, 2)} \times 7^{\max(1, 0)} \times 11^{\max(0, 1)}$$

$$= 2^3 \times 3^2 \times 5^2 \times 7^1 \times 11^1 = 8 \times 9 \times 25 \times 7 \times 11$$

$$= 138600$$

⑥

Find the numbers of distinct solutions of the linear congruence
 $15x \equiv 25 \pmod{35}$.

Sol:-

Now the form $ax \equiv b \pmod{m}$.

Now, $a=15, b=25, m=35$.

If $\gcd(a, m) = \gcd(15, 35)$.

$$\begin{array}{r} 15 \overline{) 35} \quad 2 \\ \underline{30} \\ 5 \overline{) 15} \quad 3 \\ \underline{15} \\ 0 \end{array}$$

Now, $\gcd(15, 35) = 5 = d$.

Now, $d/b = 5/25 = 5 > 1$.

$$\begin{array}{r} 5 \overline{) 25} \quad 5 \\ \underline{25} \\ 0 \end{array}$$

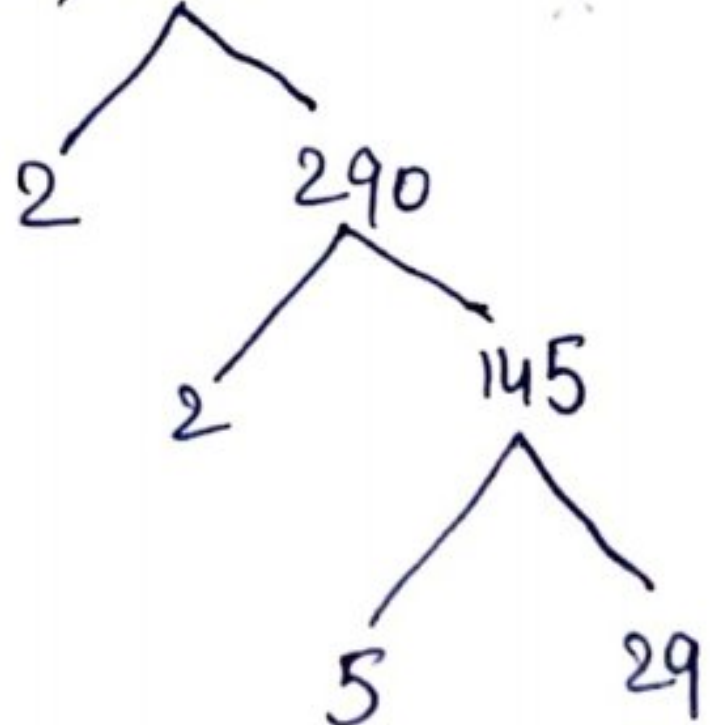
$\therefore$ The given congruence has five distinct congruence solutions.

⑦

Find the GCD and LCM of $(580, 8316)$ using prime factorization?

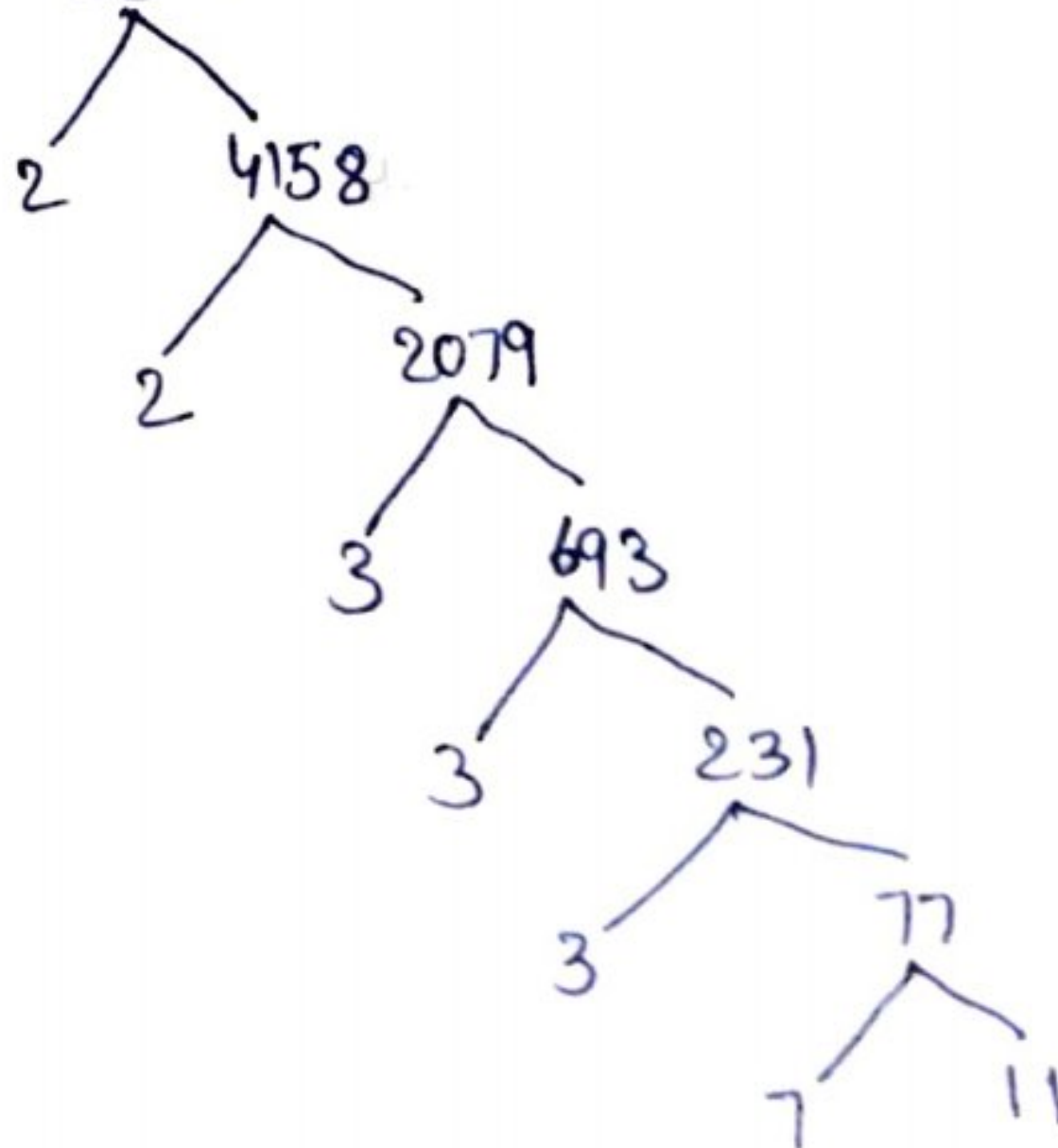
Sol:-

Now, 580



$$580 = 2^2 \times 5^1 \times 29^1$$

8316



$$8316 = 2^2 \times 3^3 \times 7 \times 11$$

Prime Factorisation of GCD Formula:-

$$\text{GCD}(a, b) = p_1^{\min(a, b)} \times p_2^{\min(a, b)} \times \dots$$

$$\begin{aligned} \text{GCD}(580, 8316) &= 2^{\min(2, 2)} \times 3^{\min(0, 3)} \times 5^{\min(0, 1)} \times 7^{\min(0, 1)} \times 11^{\min(0, 1)} \\ &\quad \times 29^{\min(0, 1)} \\ &= 2^2 \times 3^0 \times 5^0 \times 7^0 \times 11^0 \times 29^0 \\ &= 4 \times 1 \times 1 \times 1 \times 1 \times 1 = 4. \end{aligned}$$

Prime Factorisation of LCM Formula:-

$$\text{LCM}(a, b) = p_1^{\max(a, b)} \times p_2^{\max(a, b)} \times \dots$$

$$\begin{aligned} \text{LCM}(580, 8316) &= 2^{\max(2, 2)} \times 3^{\max(0, 3)} \times 5^{\max(0, 1)} \times 7^{\max(0, 1)} \\ &\quad \times 11^{\max(0, 1)} \times 29^{\max(0, 1)} \\ &= 2^2 \times 3^3 \times 5^1 \times 7^1 \times 11^1 \times 29^1 \\ &= 4 \times 27 \times 5 \times 7 \times 11 \times 29 = 1205820. \end{aligned}$$

⑧ Find the GCD(414, 662) using Euclidean Algorithm?

Sol:-

Given that, GCD(414, 662)

Now by using division algorithm (or) Euclidean Algorithm.

$$\begin{array}{r}
 \text{now, } 414 \overline{) 662} (1 \\
 \underline{414} \\
 248 \\
 248 \overline{) 414} (1 \\
 \underline{248} \\
 166 \\
 166 \overline{) 248} (1 \\
 \underline{166} \\
 82 \\
 82 \overline{) 166} (2 \\
 \underline{164} \\
 2 \\
 2 \overline{) 82} (41 \\
 \underline{82} \\
 0
 \end{array}$$

$$\text{now, } 662 = 1 \times 414 + 248, \quad q_1 = 1, \quad r_1 = 248.$$

$$414 = 1 \times 248 + 166, \quad q_2 = 1, \quad r_2 = 166.$$

$$248 = 1 \times 166 + 82, \quad q_3 = 1, \quad r_3 = 82.$$

$$166 = 2 \times 82 + 2, \quad q_4 = 2, \quad r_4 = 2.$$

$$82 = 2 \times 41 + 0, \quad q_5 = 41, \quad r_5 = 0.$$

Since $r_5 = 0$, $r_4 = 2$ is the last non-zero remainder.

$$\therefore \text{GCD}(414, 662) = 2.$$

Unit III

Short Answer Questions

①

Evaluate ${}^{10}C_8$?

Soln

Given that, ${}^{10}C_8$.

The Combination Formula:- ${}^nC_r = \frac{n!}{r!(n-r)!}$ $\rightarrow$ ①

Now, $n=10, r=8$.

Substituting n -value and r -value in equation ①,

$$\begin{aligned}\text{Now, } {}^{10}C_8 &= \frac{10!}{8!(10-8)!} \\ &= \frac{10!}{8!2!} = \frac{10 \times 9 \times 8!}{8!2!} \\ &= \frac{10 \times 9}{2 \times 1} = \frac{45}{1} = 45.\end{aligned}$$

②

If ${}^nP_2 = 72$ find the value of n ?

Soln

Given that, ${}^nP_2 = 72$.

The permutation Formula:- ${}^nP_r = \frac{n!}{(n-r)!}$

Now, $n=n, r=2$.

Now, ${}^nP_2 = 72$

$$\frac{n!}{(n-2)!} = 72$$

$$[\because n! = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1]$$

$$\frac{n(n-1)(n-2)!}{(n-2)!} = 72$$

$$n(n-1) = 72$$

$$n^2 - n = 72$$

$$\text{now, } n^2 - n - 72 = 0$$

$$n^2 - 9n + 8n - 72 = 0$$

$$n(n-9) + 8(n-9) = 0$$

$$(n-9)(n+8) = 0$$

$$\begin{array}{l|l} n-9=0 & n+8=0 \\ n=9 & n=-8 \end{array}$$

③ Compute $8P_5$ and $6C_3$.

Sol:-

Compute $8P_5$:-
Given that, $8P_5$

Permutation Formula:- $nP_r = \frac{n!}{(n-r)!} \rightarrow \text{①}$

Given, $n=8, r=5$

Substituting n and r values in equation ①, we can get

$$\begin{aligned} \text{now, } 8P_5 &= \frac{8!}{(8-5)!} = \frac{8!}{3!} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3!}{3!} \\ &= 8 \times 7 \times 6 \times 5 \times 4 = 6720. \end{aligned}$$

$$\therefore {}^8P_5 = 6720$$

Compute 6C_3 :-

Given that, 6C_3 .

Combination Formula:-

$${}^nC_r = \frac{n!}{r!(n-r)!} \rightarrow (2)$$

Given, $n=6, r=3$.

Substituting n and r value in equation (2), we get.

$$\text{now, } {}^6C_3 = \frac{6!}{3!(6-3)!}$$

$$= \frac{6!}{3!3!} = \frac{6 \times 5 \times 4 \times 3!}{3!3!}$$

$$= \frac{6 \times 5 \times 4}{3 \times 2 \times 1} \quad [\because 3! = 3 \times 2 \times 1]$$

$$= 5 \times 4 = 20$$

$$\therefore {}^6C_3 = 20$$

(4) what is the remainder of $9 \mod 7$?

Sol:-

$$\text{now, } 9 \mod 7 = 2$$

$$\begin{array}{r} 7 \overline{) 9} 1 \\ \underline{7} \\ 2 \end{array}$$

UNIT - III

Long Answer Questions

- ① How many ways to construct a sequence of 5 letters in which first 3 letters are alphabets and the next 2 letters are digits. Find with (i) Repetitions allowed.
(ii) Repetitions not allowed.

Sol:- we have to construct a sequence of 5 letters.

in which first 3 are alphabets and next 2 are digits.

we have total Alphabets - 26 (A to Z)

total digits - 10 (0, 1, 2, 3, 4, 5, 6, 7, 8, 9)

(i) Repetitions allowed:- we can choose first 3 alphabets, each in 26 ways, and next two digits each in 10 ways.

So, the required 5 letters can be chosen in,

$$\underline{26} \underline{26} \underline{26} \underline{10} \underline{10}$$

$$= 26 \times 26 \times 26 \times 10 \times 10$$

$$= 1757600 \text{ ways.}$$

(ii) Repetitions not allowed:- In this case, we choose the first alphabet in 26 ways, the next alphabet in 25 ways (no repetition)

the third alphabet in 24 ways.

∴ Alphabets can choose in $26 \times 25 \times 24$ ways.

And the 2 digits, the first digit in 10 ways and second digit in 9 ways.

∴ so the digits can choose in 10×9 ways.

Hence the five letter word can be choose when the repetition is not allowed in, $\underline{26} \underline{25} \underline{24} \underline{10} \underline{9}$

$$\begin{aligned} &= 26 \times 25 \times 24 \times 10 \times 9 \\ &= 1404000 \text{ ways.} \end{aligned}$$

② How many ways 4 boys and 4 girls can be seated in a row so that boys and girls are alternative?

Sol: Given that there are 4 boys and 4 girls.

∴ Total number of people = $4 + 4 = 8$.

We have to arrange them in a row alternatively.

Case (i): when a boy sits at the first place.

We get B G B G B G B G is the possible arrangement.

Now there are four places for 4 boys, these 4 boys can be seated in $4!$ ways.

Again, there are 4 places for 4 girls.

∴ 4 girls can be seated in $4!$ ways.

Hence the number of ways in this case is $4! \times 4!$.

$$= 4 \times 3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 1$$

$$= 24 \times 24 = 576.$$

Case (ii):- when a girl sits at the first place.

The possible arrangement is of the form.

G B G B G B G B

In this case also first we arrange 4 girls in $4!$ ways.

and 4 boys in the remaining 4 places in $4!$ ways.

$\therefore$ The number of ways in this case is $= 4! \times 4!$

$$= 4 \times 3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 1$$

$$= 24 \times 24 = 576.$$

Hence the total number of ways for the arrangement

$$= 576 + 576 = 1152.$$

③ Find the number of permutations of letters of the word "ENGINEERING".

Sol:-

Given word is ENGINEERING.

Number of letters of the word $= 11 = n$.

Number of E's $= 3$

Number of N's $= 3$

Number of G's $= 2$

Number of I's $= 2$

Number of R $= 1$.

$$\text{Total number of permutations} = \frac{11!}{3! 3! 2! 2! 1!}$$

$$= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 3 \times 2 \times 1 \times 2 \times 1 \times 2 \times 1 \times 1}$$

$$= 11 \times 10 \times 9 \times 8 \times 7 \times 5 = 277200$$

(4)

Find the number of distinct arrangements of the word "ADDRESS" if both D's are together, but s's are separated.

Ans:-

Given the word "ADDRESS"

There are total 7 letters in the given word

Now consider the two D's as one letter

so there are total 6 letters in the word

As the two s's are separated the total number of ways to arrange are.

$$\frac{6!}{2!} = \frac{720}{2} = 360.$$

If we consider two s's together the total no of ways to arrange is

$$\underline{5! = 120}$$

The number of ways to arrange if both D's are together but s's are separated is

$$360 - 120 = 240.$$

∴ The total no of ways = 240 ways